NAME ______________________ DATE ______________ PERIOD ________

Unit 10 Notes: *Measures of Spread*

Measures of spread are used to describe the distribution, or spread, of the data. The range is the difference between the greatest and least data values. Quartiles are values that divide the data set into four equal parts. The median of the lower half of a set of data is the 1st quartile and the median of the upper half of a set of data is the 3rd quartile. The difference between the third quartile and the first quartile is called the Inter quartile range. (IQR)

Example 1

Determine the measures of spread for the number of votes received for student government president:

13, 20, 18, 12, 21, 2, 18, 17, 15, 10, & 14 votes.

The greatest number in the data set is 21.

The least number is 2.

The range is 21- 2 or 19 votes.

Range

To determine the quartiles, arrange the numbers in order from least to greatest:

2, 10, 12, 13, 14, 15, 17, 18, 18, 20, 21

Q1=12 Median Q3=18

IQR

The interquartile range is 18-12 or 6.

An Outlier is a data value that is either much greater or much less than the median.

Determine the range, median, first and third quartiles and interquartile range for each data set.

Number of Boxes of Popcorn Sold						
52	72	96	~~21~~	58	~~40~~	75

21, 40, 52, 58, 72, 75, 96

Q1 Med Q3

Range: 96-21 = 75

Median: 58

Quartile 1: 40

Quartile 3: 75

IQR: 75-40 = 35

Test Grades								
~~83~~	~~83~~	~~85~~	87	89	88	~~67~~	~~79~~	~~81~~

67, 79, 81, 83, 83, 85, 87, 88, 89

Q1 = 80 | Median | Q3 = 87.5

Range: 89 − 67 = 22

Median: 83

Quartile 1: 80 $\frac{(79+81)}{2}$

Quartile 3: 87.5 $\frac{(87+88)}{2}$

IQR: 87.5 − 80 = 7.5

Number of Text Messages Sent							
20	23	~~18~~	~~4~~	~~17~~	21	~~15~~	56

4, 15, 17, 18 | 20, 21, 23, 56

Q1 = 16 | Median | Q3 = 22

Range: 56 − 20 = 36

Median: 19 $\frac{(18+20)}{2}$

Quartile 1: 16

Quartile 3: 22

IQR: 22 − 16 = 6

Raul and Bryan recorded the number of hours they slept last week. The data is shown below.

Raul's Sleep Log						
8	8.5	9.5	9	8.5	7	8.5

7, 8, 8.5, 8.5, 8.5, 9, 9.5

Q1 M Q3

IQR: 9 − 8 = 1

Bryan's Sleep Log						
6.5	7	7.5	6.5	12	7	7.5

6.5, 6.5, 7, 7, 7.5, 7.5, 12

Q1 M Q3

IQR: 7.5 − 6.5 = 1

1. Which statement is NOT true about the data provided?

   A. Both sets of data have an interquartile range of 1. ✓

   B. Both measures of center for Raul's data have a value of 8.5. False

   C. The shapes of both data distributions are non-symmetric. ✓

   D. The IQR of Bryan's data is best used to describe the spread because of the outlier. ✓

$15 + 17 + 9 + 18 = 59$

$59.00 \div 7 = 8.4$; -56, 30, -28, 20

2. Which is a true comparison of the measures of center for Bryan's data?

   ~~A.~~ Range = Mean $1 \neq 7.7$

   ~~B.~~ Mean < Median $7.7 < 7$

   C. Median > IQR $7 > 1$ True

   ~~D.~~ Mean < Median

$13 + 14 + 15 + 12 = 54$

$54.00 \div 7 = 7.71\ldots$; -49, 50, -49, 10, -7, 3